

Hugo HADJUR

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Academic Page



Google Scholar



LinkedIn

Since 2023, I have worked in applied **biostatistics and AI research** for **pharma and biotech**.
I **write peer-reviewed scientific papers**. I have a **PhD** in informatics with **ENS de Lyon**
with strong dominance in **AI**, and **NLP** experience.

WORK EXPERIENCE

SINCE DEC. 2023
Permanent



AI & STATISTICAL METHODOLOGIST AT **Saryga** (FRANCE)

— **Tech projects:** semantic analysis using LLMs, text summarization, zero-shot text classification, paraphrase mining, development of R Shiny apps and R packages, geodata analysis (supervision)

— **Biostatistics and Pharma projects:** predictive biomarker cutoff selection methods (paper is submitted), survival analysis for PDX, quantitative decision-making methods, implementation and simulation of PoC multi-arm trials, sample size optimization

SEPT. 2020 - AUG. 2023
3-year contract



PHD CANDIDATE AND ASSISTANT PROFESSOR AT **École normale supérieure de Lyon** AND **aivancity** (FRANCE)

— **PhD Thesis:** Designing sustainable, autonomous, and energy efficient Internet of Things systems, applied to smart beehives

Advisorship of **Laurent Lefèvre** & **Doreid Ammar**, at **LIP-Avalon** laboratory

— **Teaching:** advanced data analysis and visualization, data preprocessing and visualization and coding for AI and data science (Python, Tableau)

— **Intern supervision:** connected beehives deployment & data analysis

— **Head of online course platform** **aivancityX**

SEP. 2019 - AUG. 2020
& OCT. 2018 - DEC. 2018
1-year contract



INSTRUCTOR & RESEARCHER AT **emlyon Business School** (FRANCE)

— Research: Connected beehives, bees' data analysis (**IoT, machine learning, computer vision**)

— Teaching: **Python programming bootcamp** and **business analyst toolbelt** (SQL, Excel, Tableau)

JAN. 2019 - SEP. 2019
8-month contract



DATA ANALYST AT **Schneider Electric**, HOUSTON, TX (USA)

— Supported the business by building **reports/dashboards** and **KPIs** to monitor performance, conducted **database integration**

— **Communicated findings** and insights to cross-functional team members and management

— Tool used: **SQL, Tableau, Python**

OCT. 2017 - SEP. 2018
1-year exchange



STUDENT RESEARCHER AT **Kyoto University's Ishii Lab** (JAPAN)

— **Master's thesis:** master board games thanks to **Reinforcement Learning** (Q-Learning, SARSA, Deep Reinforcement Learning)

MAY-SEP. 2017
4-month internship



DATA SCIENCE RESEARCH INTERNSHIP AT **CiNet** (Center for Information and Neural Networks, Osaka, JAPAN)

— **Conducted research** about the links between **Twitter data**, human behavior and brain data, resulting in a **published peer-reviewed paper**

— **Analyzed large data sets** ($\approx 500,000$ Twitter users) thanks to **PCA, clustering & regression** techniques

— Worked in a **Japanese environment**

PUBLICATIONS

[under review] **Hugo Hadjur**, Julia Geronimi, Sandrine Guilleminot, Alessandra Serra, Libby Daniells, Marie-Karelle Riviere, Gaëlle Saint-Hilary, Pavel Mozgunov. **Evaluating methods for biomarker cut-off detection: a comparative study in small and unbalanced samples**, 2025-26

Hugo Hadjur, Doreid Ammar, Laurent Lefèvre. **Deep Reinforcement Learning for Energy-efficient Selection of Embedded Services at the Edge**, 2024 IEEE International Conferences on Internet of Things (iThings), Copenhagen, Denmark, 2024, pp. 67-74, [10.1109/iThings-GreenCom-CPSCoM-SmartData-Cybermatics62450.2024.00034](https://doi.org/10.1109/iThings-GreenCom-CPSCoM-SmartData-Cybermatics62450.2024.00034). [hal-04708697](https://hal.archives-ouvertes.fr/hal-04708697)

Hugo Hadjur, Doreid Ammar, Laurent Lefèvre. **Services Orchestration at the Edge and in the Cloud for Energy-Aware Precision Beekeeping Systems**, 2023 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW), St. Petersburg, FL, USA, 2023, pp. 769-776, [10.1109/IPDPSW59300.2023.00129](https://doi.org/10.1109/IPDPSW59300.2023.00129). [hal-04091575](https://hal.archives-ouvertes.fr/hal-04091575)

Hugo Hadjur, Doreid Ammar, Laurent Lefèvre. **Toward an intelligent and efficient beehive: A survey of precision beekeeping systems and services**, Computers and Electronics in Agriculture, Elsevier, 2022, 192, pp. 1-16. [10.1016/j.compag.2021.106604](https://doi.org/10.1016/j.compag.2021.106604). [hal-03483914v2](https://hal.archives-ouvertes.fr/hal-03483914v2)

Hugo Hadjur, Doreid Ammar, Laurent Lefèvre. **Analysis of Energy Consumption in a Precision Beekeeping System**. IoT '20 - 10th International Conference on the Internet of Things, Oct 2020, Malmö, Sweden. pp. 1-11, [10.1145/3410992.3411010](https://doi.org/10.1145/3410992.3411010). [hal-02973772](https://hal.archives-ouvertes.fr/hal-02973772)

Kazuma Mori, **Hugo Hadjur**, Masahiko Haruno. **Natural Language Content Mediates the Association Between Active Interactions on Social Network Services and Subjective Well-Being**, Cyberpsychology, Behavior, and Social Networking, 2022, pp. 678-685. [10.1089/cyber.2021.0340](https://doi.org/10.1089/cyber.2021.0340)

Alessandra Serra, Julia Geronimi, Sandrine Guilleminot, **Hugo Hadjur**, Marie-Karelle Riviere, Gaëlle Saint-Hilary, Pavel Mozgunov. **A Novel Approach to Assess the Predictiveness of a Continuous Biomarker in Early Phases of Drug Development**, Statistics in Medicine, 2025, 44: e70026. [10.1002/sim.70026](https://doi.org/10.1002/sim.70026)

COMMUNICATION

POSTERS

— **SnB** (Statistics and Biopharmacy Conference) - Paris, France, 2025. [Evaluating methods for biomarker cutoff detection: a comparative study in small and unbalanced samples](#)

— **ISCB** (Annual Conference of the International Society for Clinical Biostatistics) - Thessaloniki, Greece, 2024. [Simple Approaches for Portfolio Quantitative Decision-Making](#)

PRESENTATIONS

— [accepted] **PSI 2026** (Statisticians in the Pharmaceutical Industry Conference) - Belfast, Northern Ireland, 2026. Evaluating methods for biomarker cut-off detection

— **GreenDays 2023** - Lyon, France, 2023. Optimize task selection under energy constraints in IoT systems thanks to reinforcement learning

— **Journées du GDR Réseaux et Systèmes Distribués** - Rennes, France, 2022. Designing connected, sustainable, autonomous and low-consumption systems for a distributed network of smart beehives

— **IoT'20** - Online, 2020. Analysis of Energy Consumption in a Precision Beekeeping System

PROGRAMMING & SOFTWARE

PROGRAMMING:	Python (PyTorch, Hugging Face Transformers, NumPy, Pandas, scikit-learn, Natural Language Toolkit NLTK), R, SQL
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TECH EXPERIENCE:	Large Language Models (LLMs), Transformers, Natural Language Processing (NLP), Reinforcement Learning (RL), Computer Vision, Diffusion Models, Model Optimization
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DESIGN & TEXT:	L^AT_EX, Adobe Photoshop, Microsoft Office, Inkscape
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2020-2023		PhD at ENS de Lyon (LIP-Avalon laboratory) & aivancity Develop and model Internet of Things (IoT) systems and services based on Artificial Intelligence (AI) to support beekeepers in their work and preserve bees while studying the efficiency of connected beehives that are autonomous, sustainable, distributed and operate under limited energy budget.
2017-2018		Exchange year at Kyoto University's Graduate School of Informatics Alongside courses in English and Japanese, I worked toward my Master's thesis. This thesis proposes a reinforcement learning method applied to the game of bridge. The Q-Learning and SARSA techniques reach a level of play comparable to an intermediate human player. — Classes in English & Japanese in the department of social informatics: statistics, machine learning
2015-2017		French “Grandes Ecoles”: Ensimag, Grenoble Institute of Technology (equivalent: Master's Degree) — MMIS specialization: programming, advanced statistics, data mining, database manipulation
2013-2015		Scientific preparatory classes at CPP Grenoble (equivalent: Bachelor's Degree) — 2-year intensive university-level preparatory course to enter highly competitive Ivy League engineering schools
2007-2013	 	Middle school and high school in Tokyo (4 years at LFIT) and in Singapore (1 year at LFS) — Scientific baccalaureate with highest honours (French school)